

Glomerulonephritis Associated With Complement Disorders

Joshua M. Thurman, H. Terence Cook, Matthew C. Pickering

Glomerular diseases associated with abnormalities of the complement system include thrombotic microangiopathy (see [Chapter 30](#)) and glomerulonephritides. Activation of complement in most cases of glomerulonephritis is secondary to deposition of immune complexes. Rarely, a genetic or acquired abnormality of the complement system itself causes glomerulonephritis. Deficiencies of early components of the classic pathway of complement are associated with autoimmunity and glomerulonephritis.¹ This is most clearly seen in the very rare individuals with C1q deficiency, almost all of whom have a lupus-like illness. This is thought to be because of the failure of clearance of immunogenic apoptotic bodies and other cellular debris in the absence of normal classic pathway complement activation. The largest group of patients with complement abnormalities and glomerulonephritis (C3 glomerulopathy) exhibits abnormal control of the alternative pathway of complement activation ([Box 23.1](#)).

C3 GLOMERULOPATHY

C3 glomerulopathy encompasses glomerular disease characterized by the predominant accumulation of complement component C3 in glomeruli particularly associated with abnormal control of the alternative pathway of complement activation ([Fig. 23.1](#)).² Glomeruli show strong staining for C3 with much less intense or absent staining for immunoglobulins or for components of the classic pathway of complement activation, C1q and C4. C3 glomerulopathy is distinct from atypical hemolytic uremic syndrome (aHUS; see [Chapter 30](#)), which is also associated with alternative pathway activation, because in aHUS, complement activation is on the kidney endothelium and is not associated with well-defined deposits on electron microscopy (EM). On light microscopy, C3 glomerulopathy may manifest with mesangial proliferation, a membranoproliferative pattern (see [Chapter 22](#)), endocapillary proliferation, and crescents. Many cases that were previously classified morphologically as membranoproliferative glomerulonephritis (MPGN) types I, II, or III are actually cases of C3 glomerulopathy. Indeed, it appears that most cases that have previously been called MPGN type III (see [Chapter 22](#)) are examples of C3 glomerulopathy. However, monoclonal gammopathy-associated MPGN also may be misclassified as C3 glomerulopathy if the glomerular immunoglobulin deposits are masked and only become detectable after protease digestion of the histologic section (see later and [Chapter 22](#)).

On EM, C3 glomerulopathy also may have a variety of appearances. Common is dense deposit disease (DDD), which is characterized by replacement of the glomerular basement membrane (GBM) by dense bands on EM ([Fig. 23.2](#)). In some cases, the light microscopic appearance in DDD shows a membranoproliferative pattern, which explains the older term MPGN type II (see [Chapter 22](#)). However, most cases of DDD do not have MPGN morphology on light microscopy.

Cases of C3 glomerulopathy that do not manifest as DDD may show a range of appearances on EM with deposits that may be mesangial, subendothelial, or subepithelial and may be more or less well defined ([Fig. 23.3](#)). The relevance of the site of the deposits has not been defined, although it is likely, by analogy with diseases such as lupus nephritis, that capillary wall deposits are associated with more glomerular inflammation and higher levels of proteinuria. These cases of non-DDD C3 glomerulopathy have been given the collective name of *C3 glomerulonephritis* (C3GN).^{3,4}

Etiology and Pathogenesis

C3 glomerulopathy involves dysregulation of the alternative pathway of complement (see [Fig. 23.1](#)). In health, the alternative pathway is constantly activated at a very low rate. This means there is a constant generation of activated C3 (C3b), which allows the pathway to be rapidly amplified when needed. In the presence of pathogens, C3b amplification occurs through a positive feedback loop (C3b amplification loop) that generates millions of C3b molecules within minutes. Because amplification can progress rapidly, efficient systems are needed to prevent inappropriate activation. The most important regulator of the alternative pathway is factor H (CFH). CFH has three key functions: (1) to block the formation of alternative pathway C3 convertases by binding to C3b and inhibiting interaction between C3b and factor B, (2) to promote the spontaneous dissociation of these convertases, and (3) to work together with factor I to cleave C3b to iC3b. Mice genetically engineered to lack factor H have undetectable circulating C3 because C3 is constantly consumed by the uncontrolled alternative pathway.⁵

CFH is a glycoprotein predominantly made in the liver composed of protein subunits known as short consensus repeat (SCR) domains. The activity of CFH can be modulated by a group of closely related proteins called factor H-related (CFHR) proteins.⁶ There are five CFHR proteins in humans (CFHR1–5) encoded by individual genes downstream from the *CFH* gene. The CFHR proteins, like CFH, are composed of SCR domains and share considerable sequence similarity with CFH. This has led to genomic rearrangements (i.e., polymorphisms and mutations) within the *CFH*-*CFHR* locus, most commonly a combined deletion of the *CFHR1* and *CFHR3* genes. This is present in homozygosity in 5% to 20% of healthy individuals depending on ethnic origin. It is now clear that some CFHR proteins are able to compete with the binding of CFH to C3b.⁷ Unlike CFH, the CFHR proteins cannot inhibit complement activation. Furthermore, the CFHR-C3b interaction prevents CFH from negatively regulating C3b production. Consequently, the CFHR-C3b interaction promotes C3b amplification, termed CFH deregulation ([Fig. 23.4](#)).

In many cases of C3 glomerulopathy, the failure of CFH to control the activation of the alternative pathway in the circulation is associated with low circulating C3 because of uncontrolled consumption.

Up to 80% of patients with DDD and up to half of patients with C3GN have low serum C3.³ Many of these patients have a C3 nephritic factor (C3Nef). C3Nefs are autoantibodies that stabilize the alternative pathway C3 convertase, preventing CFH regulation. C3Nef likely plays an etiologic role in these patients. C3Nefs can be identified in 40% to 60% of cases of C3GN and 80% to 90% of cases of DDD.^{3,8} However, C3Nefs also may be found in patients with immune complex-associated MPGN, lupus nephritis, and poststreptococcal glomerulonephritis

BOX 23.1 Definitions

C3 glomerulopathy—A *disease process* secondary to abnormal control of complement activation, deposition, or degradation characterized by predominant glomerular C3 fragment deposition with electron-dense deposits on electron microscopy.

Dense deposit disease—A form of C3 glomerulopathy with a characteristic electron microscopy appearance of intensely osmiophilic transformation of the glomerular basement membrane.

C3 glomerulonephritis—C3 glomerulopathy without the characteristic appearances of dense deposit disease.

Glomerulonephritis with dominant C3—A morphologic term for cases of glomerulonephritis with dominant staining for C3c. Dominant is defined as C3c intensity 2 or more orders of magnitude more than any other immune reactant on a scale of 0 to 3 (including 0, trace, 1+, 2+, 3+). Many, but not all, of these will represent cases of C3 glomerulopathy.

and even in healthy patients. Nephritic factors that stabilize the classical pathway C3 convertase (C4Nef) or the C5 convertase (nephritic factor of the terminal pathway, or Nft) have also been detected in some patients.⁹ In some patients, failure of alternative pathway control is associated with autoantibodies directed against factor H that target its regulatory domain.

In other cases, genetic mutations lead to failure of alternative pathway control. These cases include patients with complete CFH deficiency,^{10,11} mutations in CFH that interfere with its binding to C3b,^{12,13} and mutations in C3 that change its structure so that it cannot be inhibited by CFH.¹⁴ In cases of C3 glomerulopathy without activation of circulating C3, it can be assumed that there is a failure to control the alternative pathway locally within the glomerulus. This might be because of a failure to control activation or inappropriate handling of the fragments of C3 generated by alternative pathway activation. There is a familial C3 glomerulopathy (now called CFHR5 nephropathy) without systemic C3 activation. This disease is endemic in Cyprus, and the mutation is a duplication of the first 2 exons of the *CFHR5* gene.¹⁵ This leads to an abnormal protein that forms multimers that are able to deregulate the activity of CFH on surfaces.⁷ It appears that the abnormal protein interferes with the action of CFH locally within the glomerulus and enhances alternative pathway activation.

Several other mutations and polymorphisms in complement genes have been associated with C3 glomerulopathy, but their role is uncertain. Complement mutations are only detected in a minority of patients,¹⁶ and the frequency of rare variants in complement genes was

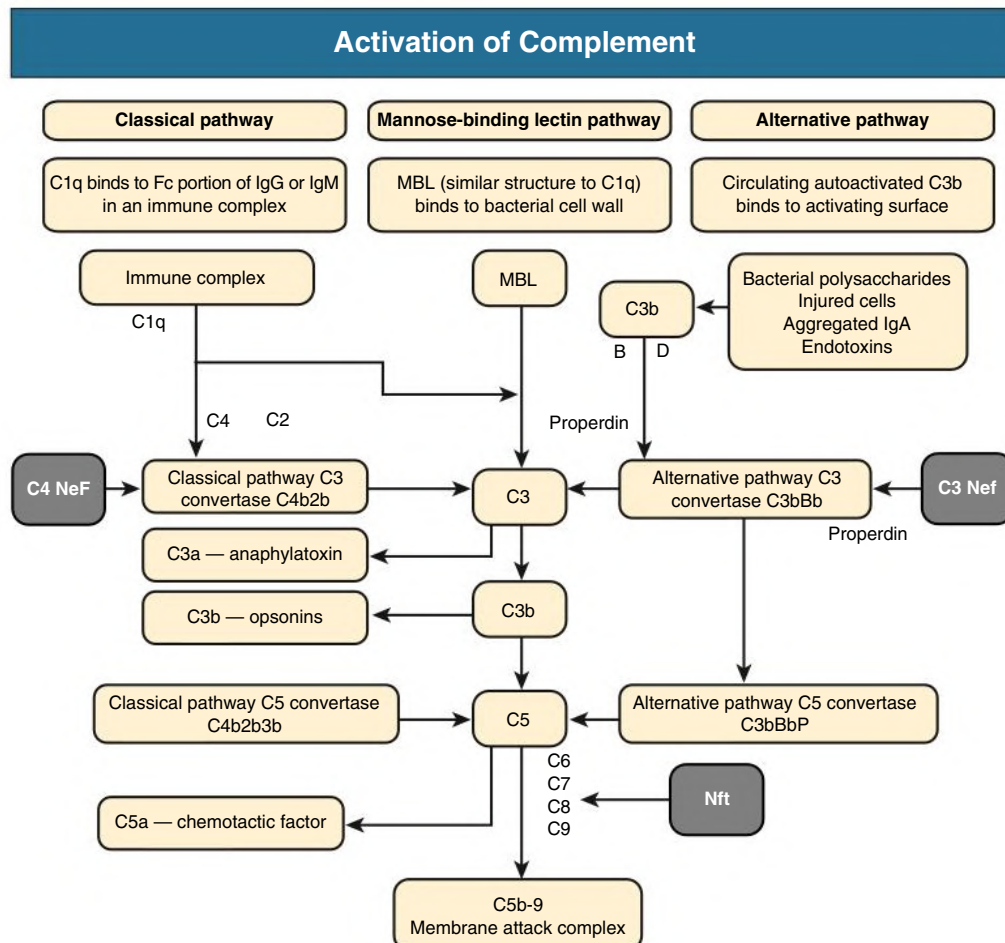


Fig. 23.1 Mechanisms of Activation of the Complement Pathways. These activators include nephritic factors (NeF, Nef). *MBL*, Mannose-binding lectin; *Nft*, Nephritic factor of the terminal pathway.

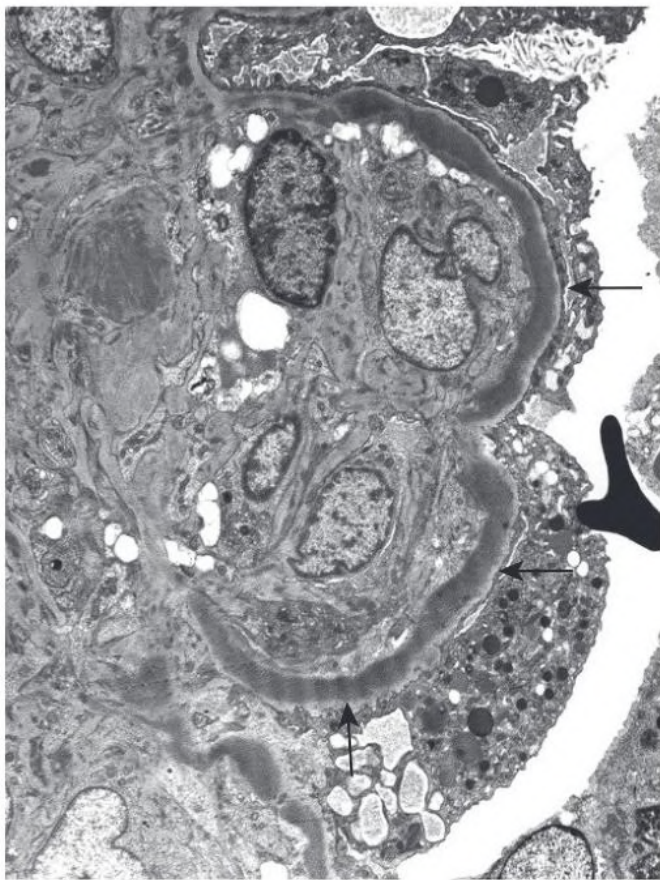


Fig. 23.2 Electron micrograph of a glomerulus showing typical appearances of dense deposit disease with very osmiophilic transformation of the basement membrane (arrows).

no higher in MPGN or C3G patients than in controls.¹⁷ There appears to be an association between monoclonal gammopathy and C3 glomerulopathy, particularly in patients who are older than 50 years.^{18–20} In some cases, the monoclonal immunoglobulin may act as a C3Nef, but this has only rarely been demonstrated.^{21,22}

Epidemiology

DDD has been reported to have a prevalence of 2 to 3 per million population and to be primarily a disease of children and young adults. However, in a 2009 series from New York, 39% of the adult patients were older than 60 years.²³ Some, but not all, studies have shown a female DDD predominance. In a French series, the ratio of C3GN to DDD was approximately 2:1 and patients with C3GN were significantly older, with a mean age at diagnosis of 30 years.³ In the United Kingdom and Ireland, the ratio was approximately 3:1 and the incidence of C3 glomerulopathy 1 to 2 per million population per year.²⁴ The apparent incidence of C3GN may increase as the entity becomes better recognized. There is no reliable information on geographic variation in incidence, with the notable exception of C3GN resulting from a specific mutation in *CFHR5*, apparently originating in Cyprus several hundred years ago.¹⁵

Clinical Manifestations

Dense Deposit Disease

At presentation, almost all patients have proteinuria, usually with hematuria. Nephrotic range proteinuria is present in two-thirds of patients^{23,25} and frank nephrotic syndrome in 12% to 65% in different series. In 98 patients from North America,²⁶ about 20% of patients did not suspect a problem and kidney disease was detected as part of a routine examination. Many patients have initial signs and symptoms

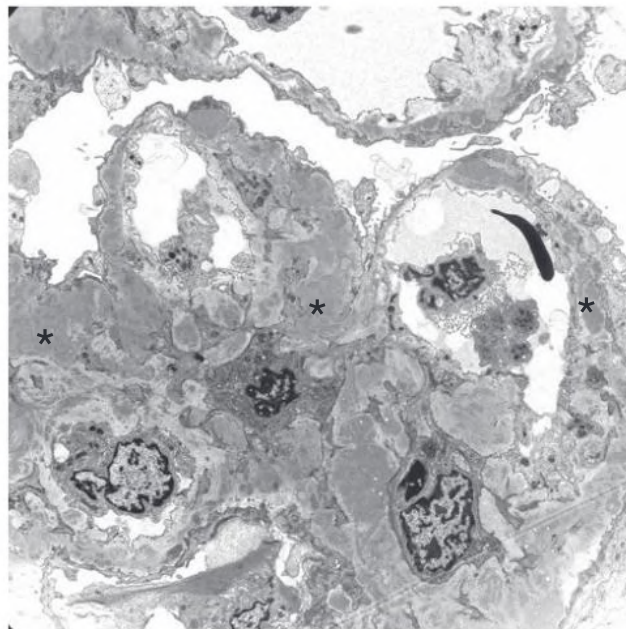


Fig. 23.3 Electron micrograph of a glomerulus in a case of C3 glomerulonephritis. There is a complex pattern of thickening of the glomerular basement membrane with intramembranous electron-dense material. Similar deposits are also seen in the mesangium (asterisks).

of acute nephritic syndrome. In some cases, there may be episodes of acute kidney injury that show complete clinical resolution.²⁷ Decreased kidney function is common at presentation, particularly in adults. Hypertension is commonly found either at presentation or in follow-up. In about half of patients, clinical onset of DDD is preceded by acute infection.

Patients with DDD may develop ocular drusen (Fig. 23.5), lipoproteinaceous deposits of complement-containing debris within the Bruch membrane beneath the retinal pigment epithelium. This pathology is similar to that in age-related macular degeneration (AMD), but, in contrast to AMD, drusen in DDD may be found as early as the second decade of life. Complete visual loss is rare, but in late disease both peripheral and central vision can be affected.²⁸ There is no correlation between the severity of the disease in the kidney and that in the eye. A small minority of patients with DDD have acquired partial lipodystrophy (APL), a condition with symmetric loss of adipose tissue from the face, arms, and upper portions of the trunk (see Fig. 23.5).

The overall long-term outcome in DDD is poor and 50% of the patients progress to end-stage kidney disease (ESKD) within 10 years of diagnosis; young women appear to be at especially high risk.²⁶

C3 Glomerulonephritis

C3GN has only recently been recognized, and thus clinical manifestations are less well defined. In a French series, 27% of patients with C3GN had nephrotic syndrome, two-thirds had microhematuria, and one-third had hypertension at presentation.³ In that series, the rate of progression to ESKD was similar to that in the patients with DDD.

In *CFHR5* nephropathy in Cyprus, the major clinical feature in young patients is microhematuria, which is present in 90%, and 20% of patients reported episodes of macrohematuria often associated with upper respiratory tract infection.²⁹ Proteinuria became more common with increasing age and was seen in 80% of males and 20% of females over the age of 50 years. Impaired kidney function was more common with increasing age, particularly in men, and of 18 patients who reached ESKD, 78% were male.

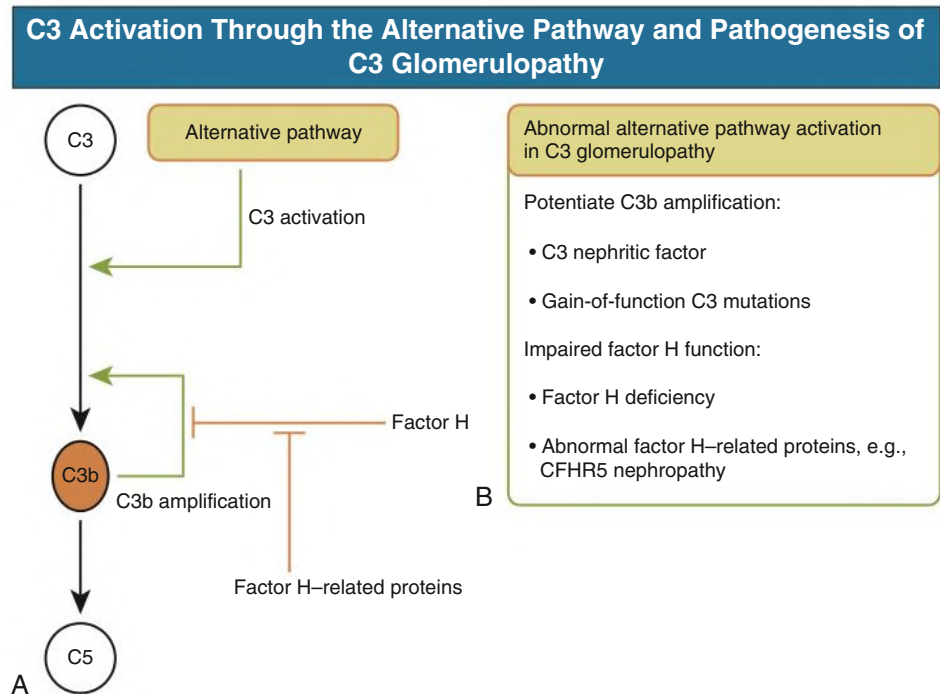


Fig. 23.4 C3 Activation through the Alternative Pathway and Pathogenesis of C3 Glomerulopathy. (A) Activation of C3 results in the formation of C3b and then C5 activation. C3b production is amplified through a positive feedback loop. Factor H inhibits this amplification (depicted). Some factor H-related proteins (e.g., complement factor H-related protein 5 [CFHR5]) interact with C3b and, unlike factor H, allow C3b amplification to proceed. In this way, they antagonize the actions of factor H in a process termed factor H deregulation. (B) C3 glomerulopathy is associated with abnormal (increased) activation of the alternative pathway. These include situations in which there is enhanced C3b amplification despite normal factor H function, such as in the presence of C3 nephritic factor, and where there is a defect in the ability of factor H to negatively regulate the alternative pathway. Examples include cases of factor H deficiency (extremely rare) and abnormal factor H-related proteins associated with familial C3 glomerulonephritis (e.g., CFHR5 nephropathy).

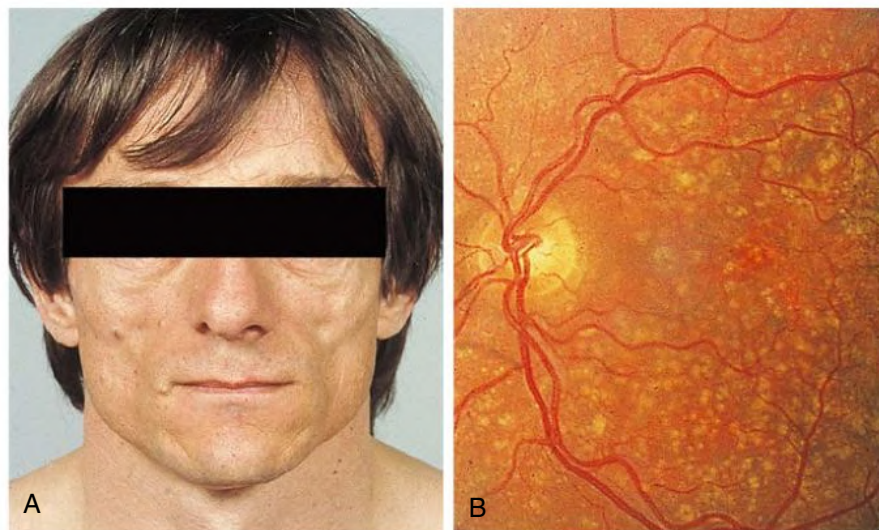


Fig. 23.5 Dense Deposit Disease. (A) Partial lipodystrophy. Note the absence of subcutaneous fat from the face. (B) Drusen bodies in the retina. (Courtesy Dr. C.D. Short, Manchester, UK.)

Laboratory Findings

Low levels of serum C3 are found in approximately 80% of patients with DDD and up to 50% of patients with C3GN. In C3GN secondary to CFHR5 nephropathy, serum C3 levels are typically normal. Serum levels of the early components of the classic pathway (C1q and C4) are usually normal. Most patients with DDD are positive for serum C3Nef, and in

more than 50% of patients C3Nef persists throughout the clinical course.³⁰ However, C3Nef is not a specific serologic marker (see earlier). C3Nef can be identified in 40% to 60% of cases of C3GN.^{3,8} However, methods for measuring C3Nef and the other nephritic factors are not standardized.

Box 23.2 shows a list of investigations that may be helpful in C3 glomerulopathy.^{31,32} It is recommended that investigations in the

BOX 23.2 Serologic and Genetic Evaluation of C3 Glomerulopathy

Tests Recommended in All Patients With C3 Glomerulopathy

- Measurement of serum levels of C3 and C4
- Measurement of C3 nephritic factor
- CH50—classic pathway hemolytic assay, and AH50—alternative pathway hemolytic assay
- Measurement of factor H and factor I
- Serum paraprotein detection
- Testing for the genetic mutation of CFHR5 nephropathy
- Anti-factor H autoantibodies

Tests That Should Be Considered on a Case-by-Case Basis

- Measurement of serum factor B and C5
- Measurement of markers of C3 and C5 activation (e.g., C3d, Bb, soluble C5b9)
- Detection of autoantibodies to factor B
- Mutation testing of complement regulatory genes (e.g., CFH, CFI, CD46), activation protein genes (*C3*, *CFB*), and assessment of copy number variation across the *CFH-CFHR* locus

second category be discussed with experts and performed in laboratories with experience in complement assays. Complement laboratories in Europe are listed on the European Complement Network website (www.eocomplement.org). Parameter selection should consider the clinical scenario. For example, in patients with C3 glomerulopathy and low serum C3 levels in the absence of C3Nef, it is important to test for anti-CFH autoantibodies. In familial cases of C3 glomerulopathy, a search for genetic mutations may be important. It is important to test for a circulating paraprotein in adult patients. C3 glomerulopathy patients who have a detectable paraprotein in the absence of an underlying malignancy are classified as having a monoclonal gammopathy of renal significance (MGRS).³³

Pathology

The defining feature of C3 glomerulopathy is the presence of C3 (usually detected with an antibody to C3c) in glomeruli by immunohistology (Fig. 23.6). In most cases, C3 is seen on capillary walls and in the mesangium. In some cases, particularly in DDD, C3 also may be found on Bowman's capsule or on tubular basement membranes.

The light microscopic appearances are quite variable. In both DDD and C3GN, membranoproliferative changes are common with increased glomerular lobulation, an increase in mesangial matrix and cells, and capillary wall thickening with double contour formation (Fig. 23.7). In patients with C3GN, 71% showed an MPGN pattern.³ Other cases may exhibit a mesangial proliferative pattern or endocapillary hypercellularity, in part as a result of influx of macrophages or neutrophils. Sometimes this endocapillary hypercellularity may affect almost all the glomeruli, giving an appearance of diffuse endocapillary proliferative glomerulonephritis similar to that typically seen with postinfectious glomerulonephritis (PIGN). Crescent formation may be sufficiently prominent to merit the designation of crescentic glomerulonephritis (>50% crescents). In DDD, the distribution of different histologic patterns was membranoproliferative (25%), mesangial proliferative (45%), crescentic (18%), and acute proliferative and exudative (12%).³⁴

The EM appearances of C3 glomerulopathy are also variable, but in many cases the diagnosis of C3 glomerulopathy can be suspected from the EM changes. By definition, DDD shows the presence of osmophilic dense transformation of the GBM (see Fig. 23.2), with similar

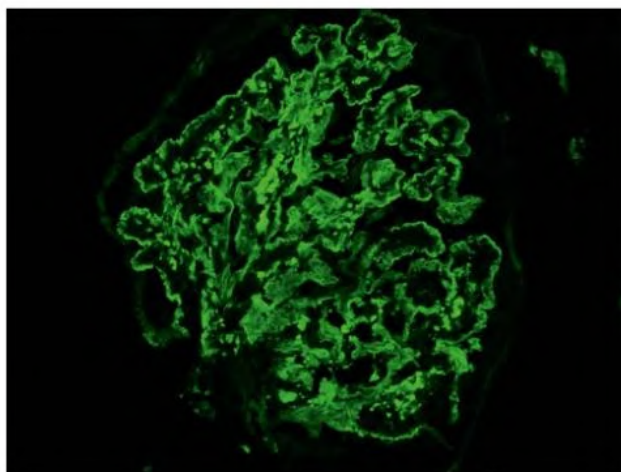


Fig. 23.6 Immunofluorescence for C3c in a case of dense deposit disease. There is widespread staining of capillary walls and focal granular mesangial staining.

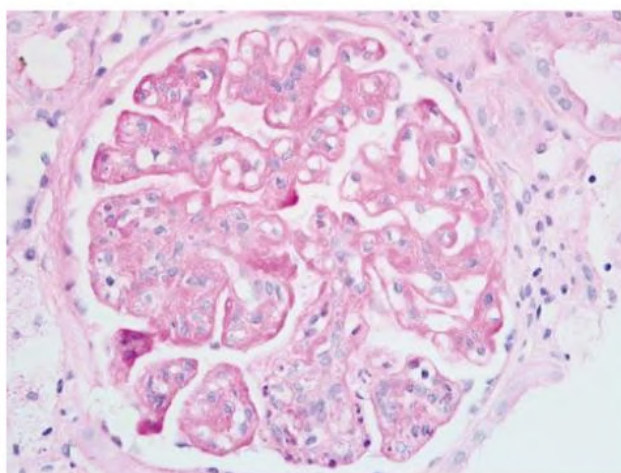


Fig. 23.7 C3 Glomerulonephritis. Light microscopy in a case of C3 glomerulonephritis. The glomerulus shows a membranoproliferative pattern of injury, with increased mesangium, capillary wall thickening, and segmental endocapillary hypercellularity (PAS stain).

features often seen in Bowman's capsule and tubular basement membranes. These changes may be segmental within glomeruli, in some cases making it difficult to define DDD with certainty. In DDD, there are typically large electron densities in the mesangium.

In some cases of C3GN, electron-dense material expands the GBM, similar to the changes in DDD but without such marked electron density; the distinction between these cases and DDD may not be clear cut and are subjective (see Fig. 23.3).³⁵ Other cases have more distinct subendothelial and mesangial electron-dense deposits reminiscent of those seen in immune complex MPGN. Some cases show a complex pattern of intramembranous deposits that was previously designated as a form of MPGN type III (see Fig. 23.3).

In both DDD and C3GN, subepithelial hump-shaped deposits, identical to those characteristically seen in PIGN, occur. Their significance is unclear, although they may be more common in infectious exacerbations of the disease.

In one study the presence of crescents was associated with adverse kidney outcomes.¹⁶ In that study, activity and chronicity scores (analogous to those used in lupus nephritis) also correlated with kidney disease progression.

Differential Diagnosis

The diagnosis of C3 glomerulopathy is relatively straightforward if there is isolated C3 deposition with typical deposits on EM. However, some patients have otherwise typical appearances of DDD or C3GN but also have small amounts of immunoglobulin in glomeruli, which presents a diagnostic challenge. The criterion with the best balance of sensitivity and specificity is the presence of dominant C3 staining with the intensity of C3 staining at least two orders of magnitude greater than any other immunoreactant (i.e., immunoglobulin [Ig] G, IgM, IgA, and C1q).^{31,36} This is referred to as glomerulonephritis with dominant C3 (see [Box 23.1](#)). Cleavage of C3c generates the C3d fragment, which remains bound within the glomerulus. Antibodies that detect C3d may be useful for diagnosing C3 glomerulopathy in biopsies where C3c deposits appear faint.³⁷ In some cases, the initial kidney biopsy may not show C3-dominant GN, but it may appear in subsequent biopsies,^{36,38,39} suggesting that repeat biopsy may be useful in cases with an atypical clinical course.

Some patients with monoclonal gammopathies have glomerular immunoglobulin deposits that are not detected by standard immunofluorescence and only after tissue digestion with pronase.^{40,41} It is recommended that screening for monoclonal gammopathy be performed in all patients presenting with C3G.

Another problem is the distinction of C3 glomerulopathy from PIGN. PIGN may show markedly reduced serum C3, and glomeruli may stain for C3 without immunoglobulin. In some cases, the distinction from C3 glomerulonephritis may be possible only by following the patient for resolution of clinical glomerulonephritis.³¹ It has been suggested that PIGN could be considered a self-limiting form of C3 glomerulopathy. Some patients who have been diagnosed with atypical PIGN have C3 glomerulopathy.⁴² Cryoglobulinemia also should be considered in the differential diagnosis because cases may present with prominent C3 with only sparse IgG staining. Anti-Factor B autoantibodies are transiently detectable in a high proportion of children with acute PIGN.⁴³ If validated, this may represent a means of distinguishing PIGN from C3G at presentation.

Treatment

The optimal treatment for C3 glomerulopathy remains undefined. In addition, in many studies DDD was grouped with forms of MPGN, which makes it difficult to make specific statements about DDD. Treatments that have been tried include renin angiotensin system

blockade, corticosteroids and other immunosuppressants, anticoagulants, and plasma exchange.⁴⁴ Several case series have found that mycophenolate mofetil (MMF) may be effective in C3 glomerulonephritis.^{45,46} Nephritic factors are a form of autoantibody, and patients with nephritic factors may be more likely to respond to treatment with immunosuppression than patients with genetic causes of complement dysregulation. Nevertheless, some patients with complement gene mutations still respond to treatment, and there are not yet any lab tests that can be used to select or exclude patients for treatment. A Kidney Disease: Improving Global Outcomes (KDIGO) controversies conference³¹ recommended that all patients should receive optimal blood pressure control and patients with moderate disease—defined as urine protein greater than 500 mg/24 h despite supportive therapy, moderate inflammation on kidney biopsy, or recent rise in creatinine—should receive prednisone or MMF. In severe disease with proteinuria more than 2 g/24 h, severe disease on biopsy, or progressive creatinine increase, the KDIGO conference suggested the use of methylprednisolone pulse dosing and other immunosuppressants.³¹

In many cases of C3 glomerulopathy, the deposition of C3 in glomeruli leads to subsequent activation of C5, and there are a number of case series reporting on use of the anti-C5 antibody eculizumab to treat C3 glomerulopathy.^{47,48} Some, but not all, patients have shown clinical improvement, and in some there was reduction of glomerular inflammation on repeat biopsy. Patients with rapidly progressive disease and glomerular crescents may be more likely to respond to treatment.⁴⁹ In the patients who underwent repeat biopsy, IgG-κ was found, consistent with binding of the monoclonal eculizumab to C5 in glomeruli.⁵⁰ However, the KDIGO conference concluded, “Data are insufficient to recommend eculizumab as a first-line agent for the treatment of rapidly progressive disease.”³¹ Rational treatment of C3 glomerulopathy would involve inhibition of C3 activation, and a number of drugs are in clinical trials that may achieve that and allow more targeted therapy in the future.

In patients with an underlying monoclonal gammopathy, treatment of the hematologic process with B-cell or plasma cell-directed therapies can lead to improvement of the kidney disease.⁵¹ Transplantation is a reasonable option for patients who progress to ESKD, particularly because so many of these patients are children. Disease recurrence is seen in the majority of patients, however, and recurrence of C3 glomerulopathy after transplantation is higher than recurrence of other causes of primary glomerulonephritis (see [Chapter 113](#)).^{52,53}

SELF-ASSESSMENT QUESTIONS

- Which of the following are *not* features of C3 glomerulopathy?
 - Subendothelial electron-dense deposits
 - Full house of immunoglobulins and complement on immunofluorescence
 - Subepithelial electron-dense humps
 - Mesangial C3 on immunofluorescence
 - Glomerular crescents
- Which of the following *best* describes C3 nephritic factor?
 - A paraprotein
 - An autoantibody against ADAMTS13
 - An autoantibody against complement factor H
 - An abnormal factor H-related protein
 - An autoantibody against the alternative pathway C3 convertase
- Which of the following *best* describes CFHR5 nephropathy?
 - C3 glomerulopathy associated with C3 nephritic factor
 - C3 glomerulopathy associated with complete absence of CFHR5
 - C3 glomerulopathy associated with intramembranous electron-dense deposits
 - C3 glomerulopathy associated with heterozygous mutation of CFHR5
 - C3 glomerulopathy associated with autoantibodies against CFHR5
- Which of the following is an evidence-based therapy for C3 glomerulopathy?
 - CFHR5 replacement
 - Alternate-day glucocorticoid treatment
 - Anti-C5 antibody therapy (eculizumab)
 - Ponticelli regimen
 - None of the above

REFERENCES

- Ballanti E, Perricone C, Greco E, et al. Complement and autoimmunity. *Immunol Res*. 2013;56(2–3):477–491.
- Fakhouri F, Fremeaux-Bacchi V, Noel LH, Cook HT, Pickering MC. C3 glomerulopathy: a new classification. *Nat Rev Nephrol*. 2010;6(8):494–499.
- Servais A, Noel LH, Roumenina LT, et al. Acquired and genetic complement abnormalities play a critical role in dense deposit disease and other C3 glomerulopathies. *Kidney Int*. 2012;82(4):454–464.
- Sethi S, Fervenza FC, Zhang Y, et al. C3 glomerulonephritis: clinicopathological findings, complement abnormalities, glomerular proteomic profile, treatment, and follow-up. *Kidney Int*. 2012;82(4):465–473.
- Pickering MC, Cook HT, Warren J, et al. Uncontrolled C3 activation causes membranoproliferative glomerulonephritis in mice deficient in complement factor H. *Nat Genet*. 2002;31:424–428.
- Jozsi M, Tortajada A, Uzonyi B, Goicoechea de Jorge E, Rodriguez de Cordoba S. Factor H-related proteins determine complement-activating surfaces. *Trends Immunol*. 2015;36(6):374–384.
- Goicoechea de Jorge E, Caesar JJ, Malik TH, et al. Dimerization of complement factor H-related proteins modulates complement activation in vivo. *Proc Natl Acad Sci U S A*. 2013;110(12):4685–4690.
- Iatropoulos P, Noris M, Mele C, et al. Complement gene variants determine the risk of immunoglobulin-associated MPGN and C3 glomerulopathy and predict long-term renal outcome. *Mol Immunol*. 2016;71:131–142.
- Zhao F, Afonso S, Lindner S, et al. C3-Glomerulopathy autoantibodies mediate distinct effects on complement C3- and C5-convertases. *Front Immunol*. 2019;10:1030.
- Levy M, Halbwachs-Mecarelli L, Gubler MC, et al. H deficiency in two brothers with atypical dense intramembranous deposit disease. *Kidney Int*. 1986;30(6):949–956.
- Dragon-Durey MA, Fremeaux-Bacchi V, Loirat C, et al. Heterozygous and homozygous factor h deficiencies associated with hemolytic uremic syndrome or membranoproliferative glomerulonephritis: report and genetic analysis of 16 cases. *J Am Soc Nephrol*. 2004;15(3):787–795.
- Licht C, Heinen S, Jozsi M, et al. Deletion of Lys224 in regulatory domain 4 of Factor H reveals a novel pathomechanism for dense deposit disease (MPGN II). *Kidney Int*. 2006;70(1):42–50.
- Wong EK, Anderson HE, Herbert AP, et al. Characterization of a factor H mutation that perturbs the alternative pathway of complement in a family with membranoproliferative GN. *J Am Soc Nephrol*. 2014;25(11):2425–2433.
- Martinez-Barricarte R, Heurich M, Valdes-Canedo F, et al. Human C3 mutation reveals a mechanism of dense deposit disease pathogenesis and provides insights into complement activation and regulation. *J Clin Invest*. 2010;120(10):3702–3712.
- Gale DP, de Jorge EG, Cook HT, et al. Identification of a mutation in complement factor H-related protein 5 in patients of Cypriot origin with glomerulonephritis. *Lancet*. 2010;376(9743):794–801.
- Bombach AS, Santoriello D, Avasare RS, et al. C3 glomerulonephritis and dense deposit disease share a similar disease course in a large United States cohort of patients with C3 glomerulopathy. *Kidney Int*. 2018;93(4):977–985.
- Levine AP, Chan MMY, Sadeghi-Alavijeh O, et al. Large-scale whole-genome sequencing reveals the genetic architecture of primary membranoproliferative GN and C3 glomerulopathy. *J Am Soc Nephrol*. 2020;31(2):365–373.
- Sethi S, Sukov WR, Zhang Y, et al. Dense deposit disease associated with monoclonal gammopathy of undetermined significance. *Am J Kidney Dis*. 2010;56(5):977–982.
- Bridoux F, Desport E, Fremeaux-Bacchi V, et al. Glomerulonephritis with isolated C3 deposits and monoclonal gammopathy: a fortuitous association? *Clin J Am Soc Nephrol*. 2011;6(9):2165–2174.
- Zand L, Kattah A, Fervenza FC, et al. C3 glomerulonephritis associated with monoclonal gammopathy: a case series. *Am J Kidney Dis*. 2013.
- Jokiranta TS, Solomon A, Pangburn MK, Zipfel PF, Meri S. Nephritogenic lambda light chain dimer: a unique human miniautoantibody against complement factor H. *J Immunol*. 1999;163(8):4590–4596.
- Meri S, Koistinen V, Miettinen A, Tornroth T, Seppala JJ. Activation of the alternative pathway of complement by monoclonal lambda light chains in membranoproliferative glomerulonephritis. *J Exp Med*. 1992;175(4):939–950.
- Nasr SH, Valeri AM, Appel GB, et al. Dense deposit disease: clinicopathologic study of 32 pediatric and adult patients. *Clin J Am Soc Nephrol*. 2009;4(1):22–32.
- Medjeral-Thomas NR, O'Shaughnessy MM, O'Regan JA, et al. C3 glomerulopathy: clinicopathologic features and predictors of outcome. *Clin J Am Soc Nephrol*. 2014;9(1):46–53.
- Little MA, Dupont P, Campbell E, Dorman A, Walshe JJ. Severity of primary MPGN, rather than MPGN type, determines renal survival and post-transplantation recurrence risk. *Kidney Int*. 2006;69(3):504–511.
- Lu DF, Moon M, Lanning LD, McCarthy AM, Smith RJ. Clinical features and outcomes of 98 children and adults with dense deposit disease. *Pediatr Nephrol*. 2012;27(5):773–781.
- Antoine B, Faye C. The clinical course associated with dense deposits in the kidney basement membranes. *Kidney Int*. 1972;1(6):420–427.
- Savigne J, Amos L, Ierino F, et al. Retinal disease in the C3 glomerulopathies and the risk of impaired vision. *Ophthalmic Genet*. 2016;37(4):369–376.
- Athanasίου Y, Voskarides K, Gale DP, et al. Familial C3 glomerulopathy associated with CFHR5 mutations: clinical characteristics of 91 patients in 16 pedigrees. *Clin J Am Soc Nephrol*. 2011;6(6):1436–1446.
- Schwartz R, Rother U, Anders D, Gretz N, Scharer K, Kirschfink M. Complement analysis in children with idiopathic membranoproliferative glomerulonephritis: a long-term follow-up. *Pediatr Allergy Immunol*. 2001;12(3):166–172.
- Goodship TH, Cook HT, Fakhouri F, et al. Atypical hemolytic uremic syndrome and C3 glomerulopathy: conclusions from a “Kidney Disease: Improving Global Outcomes” (KDIGO) controversies conference. *Kidney Int*. 2017;91(3):539–551.
- Pickering MC, D'Agati VD, Nester CM, et al. C3 glomerulopathy: consensus report. *Kidney Int*. 2013;84(6):1079–1089.
- Leung N, Bridoux F, Batuman V, et al. The evaluation of monoclonal gammopathy of renal significance: a consensus report of the International Kidney and Monoclonal Gammopathy Research Group. *Nat Rev Nephrol*. 2019;15(1):45–59.
- Walker PD, Ferrario F, Joh K, Bonsib SM. Dense deposit disease is not a membranoproliferative glomerulonephritis. *Mod Pathol*. 2007;20(6):605–616.
- Sethi S, Fervenza FC, Smith RJ, Haas M. Overlap of ultrastructural findings in C3 glomerulonephritis and dense deposit disease. *Kidney Int*. 2015;88(6):1449–1450.
- Hou J, Markowitz GS, Bombach AS, et al. Toward a working definition of C3 glomerulopathy by immunofluorescence. *Kidney Int*. 2014;85(2):450–456.
- Snijders MLH, van de Wall-Neecke BJ, Hesselink DA, Becker JU, Clahsen-van Groningen MC. Utility of immunohistochemistry with C3d in C3 glomerulopathy. *Mod Pathol*. 2020;33(3):431–439.
- Kerns E, Rozansky D, Troxell ML. Evolution of immunoglobulin deposition in C3-dominant membranoproliferative glomerulopathy. *Pediatr Nephrol*. 2013;28(11):2227–2231.
- Kersnik Levart T, Ferluga D, Vizjak A, Mraz J, Kojc N. Severe active C3 glomerulonephritis triggered by immune complexes and inactivated after eculizumab therapy. *Diagn Pathol*. 2016;11(1):94.
- Larsen CP, Messias NC, Walker PD, et al. Membranoproliferative glomerulonephritis with masked monotypic immunoglobulin deposits. *Kidney Int*. 2015;88(4):867–873.
- Cook HT. Evolving complexity of complement-related diseases: C3 glomerulopathy and atypical haemolytic uremic syndrome. *Curr Opin Nephrol Hypertens*. 2018;27(3):165–170.
- Sethi S, Fervenza FC, Zhang Y, et al. Atypical postinfectious glomerulonephritis is associated with abnormalities in the alternative pathway of complement. *Kidney Int*. 2013;83(2):293–299.
- Chauvet S, Berthaud R, Devriese M, et al. Anti-factor B antibodies and acute postinfectious GN in children. *J Am Soc Nephrol*. 2020;31(4):829–840.

44. Appel GB, Cook HT, Hageman G, et al. Membranoproliferative glomerulonephritis type II (dense deposit disease): an update. *J Am Soc Nephrol*. 2005;16(5):1392–1403.
45. Rabasco C, Caverio T, Roman E, et al. Effectiveness of mycophenolate mofetil in C3 glomerulonephritis. *Kidney Int*. 2015;88(5):1153–1160.
46. Caravaca-Fontan F, Diaz-Encarnacion MM, Lucientes L, et al. Mycophenolate mofetil in C3 glomerulopathy and pathogenic drivers of the disease. *Clin J Am Soc Nephrol*. 2020;15(9):1287–1298.
47. Bomback AS, Smith RJ, Barile GR, et al. Eculizumab for dense deposit disease and C3 glomerulonephritis. *Clin J Am Soc Nephrol*. 2012;7(5):748–756.
48. Ruggenti P, Daina E, Gennarini A, et al. C5 Convertase blockade in membranoproliferative glomerulonephritis: a Single-Arm clinical trial. *Am J Kidney Dis*. 2019;74(2):224–238.
49. Le Quintrec M, Lapeyraque AL, Lionet A, et al. Patterns of clinical response to eculizumab in patients with C3 glomerulopathy. *Am J Kidney Dis*. 2018;72(1):84–92.
50. Herlitz LC, Bomback AS, Markowitz GS, et al. Pathology after eculizumab in dense deposit disease and C3 GN. *J Am Soc Nephrol*. 2012;23(7):1229–1237.
51. Chauvet S, Fremeaux-Bacchi V, Petitprez F, et al. Treatment of B-cell disorder improves renal outcome of patients with monoclonal gammopathy-associated C3 glomerulopathy. *Blood*. 2017;129(11):1437–1447.
52. Regunathan-Shenk R, Avasare RS, Ahn W, et al. Kidney transplantation in C3 glomerulopathy: a case series. *Am J Kidney Dis*. 2019;73(3):316–323.
53. Ham YR, Song CH, Bae HJ, et al. Growth differentiation factor-15 as a predictor of idiopathic membranous nephropathy progression: a retrospective study. *Dis Markers*. 2018;2018:1463940.